



SEM SPONGE DETECTIVE MISSION!



Welcome, Microbe Detective! Your mission is to explore kitchen sponges using a Scanning Electron Microscope (SEM) simulator and report your discoveries. You'll compare three sponge samples and figure out where bacteria are hiding!

https://myscope-explore.org/virtualSEM_explore.html

1. Select the **Unused Kitchen Sponge** sample.

In the boxes below, draw this sample at three different points of magnification. Label the bottom right corner of each box with the magnification level you selected to draw.

ANY SIGNS
OF BACTERIA?

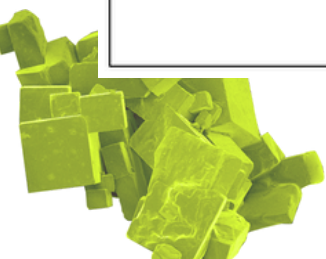
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2. Select the **Kitchen sponge with microbes 1** sample.

In the boxes below, draw this sample at three different points of magnification. Label the bottom right corner of each box with the magnification level you selected to draw.

WHAT'S
CHANGED?

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3. Select the **Kitchen sponge with microbes 2** sample.

In the boxes below, draw this sample at three different points of magnification. Label the bottom right corner of each box with the magnification level you selected to draw.

**SPOT THE
DIFFERENCE**

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FINAL CHALLENGE: MICROBE MASTERMIND

1. Why do bacteria love sponges so much?





2. What else might grow on a used sponge?





3. How could we stop bacteria from taking over our sponges?







Design a "Super Sponge" that resists bacteria! What would it look like? What materials would you use?



Draw it here!